

# HimBhasha AI

## Preserving and Empowering Himachal's Regional Languages through Artificial Intelligence

AI for Bharat Hackathon 2026 Proposal



# AI for Bharat Hackathon 2026

**Track:** Regional Language AI (Voice, Translation, Low-Resource NLP)

## Problem Statement

### Proposed Solution: HimBhasha AI

**HimBhasha AI** is envisioned as an intelligent, multilingual assistant designed to bridge this linguistic gap and empower the people of Himachal Pradesh. Our initial Minimum Viable Product (MVP) will focus on Kangdi, with a robust architecture designed for seamless expansion to other regional languages. The core features will include:

- **Text and Voice Communication:** Interact naturally in your preferred regional dialect.
- **Multilingual Translation:** Effortlessly translate between Kangdi, other Himachal regional languages, Hindi, and English.
- **Document OCR & Simplification:** Upload documents, extract text using OCR, and receive simplified explanations in your chosen language.
- **AI Question Answering:** Get accurate information and answers tailored to your regional context.
- **Natural Voice Responses:** Receive information in a clear, natural-sounding voice.

## Key Features

-  **Voice-to-Voice AI Assistant:** Natural, conversational AI interaction.
-  **Multilingual Translation:** Seamless language conversion.
-  **Document OCR & Simplification:** Extract and understand document content easily.
-  **AI Question Answering:** Context-aware information retrieval.
-  **Regional Language Learning:** Support for learning and practicing local dialects.
-  **Government Service Assistant:** Simplified access to public services.
-  **Offline-ready Architecture:** Ensuring accessibility even with limited connectivity.

# AI Workflow

## AI Workflow Flowchart

User Input (Voice/Image/Text) → Speech Recognition/OCR → Language Detection → Translation + AI Understanding → AI Response Generation → Text + Voice Output

## Technology Stack & Roadmap

### Tech Stack

Our technology stack is built for scalability, performance, and rapid development:

#### Frontend:

- React
- Next.js
- Tailwind CSS

#### Backend:

- FastAPI

#### AI Models:

- Whisper (Speech Recognition)
- OCR (PaddleOCR / Tesseract)
- Gemini API (LLM Capabilities)
- Indic NLP Models
- Text-to-Speech (TTS)

#### Database:

- Firebase / Supabase

#### Deployment:

- Vercel / Render

### Expected Impact

- **Enhanced Digital Accessibility:** Significantly improve access to digital services for speakers of Himachal's regional languages.
- **Bridged Communication Gap:** Foster better communication between local communities and government, healthcare, and educational institutions.
- **Language Preservation:** Utilize AI to document, digitize, and revitalize low-resource regional languages.
- **Scalable Multilingual Platform:** Build a robust foundation that can be expanded to support other low-resource Indian languages.

### Future Roadmap

- **Phase 1 (MVP):** Focus on **Kangdi** language support.
- **Phase 2:** Expand to **Mandeali** and **Kullui**.
- **Phase 3:** Integrate **Chambeali**, **Sirmauri**, and **Kinnauri**.
- **Phase 4:** Generalize and adapt the platform for other **low-resource Indian languages**.